

Student Utility Program

1. Aim

To develop a menu-driven Student Utility Program in Java that demonstrates basic Java concepts such as methods, parameters, return values, conditional statements, switch-case, loops, arrays, and user input using Scanner.

2. Step-by-Step Procedure

1. Create a Java class named StudentUtilityProgram.
2. Import the Scanner class to accept input from the user.
3. Create methods for different student utility operations.
4. Create methods for student information, total marks, percentage, and grade.
5. Create methods for even/odd checking, largest of three numbers, grade, and day of week.
6. Create loop-based methods for multiplication table, numbers from 1 to N, sum, and Fibonacci series.
7. Create methods with parameters and return values for factorial, prime checking, maximum, and circle area.
8. Display a menu containing all operations in main().
9. Read the user's choice and use switch-case to call the corresponding method.
10. Repeat the menu using a do-while loop until the user selects 0.
11. Close the Scanner object after the program terminates.

3. Algorithm

1. Start.
2. Create a Scanner object.
3. Display the Student Utility Program menu.
4. Read the user's choice.
5. Execute the selected operation using switch-case.
6. Display the result.
7. Repeat the menu until the choice is 0.
8. Close Scanner and stop.

4. UML Class Diagram

```
+-----+
| StudentUtilityProgram |
+-----+
| Methods:              |
| + displayInfo(name, roll, marks) |
| + calTotal(marks) : int |
| + calPercentage(total, subjects) : double |
| + evenOdd(num)         |
| + findLargest(a, b, c) |
| + displayResult(percentage) |
| + dayOfWeek(dayNumber) |
| + printTable(num)      |
| + displayNumbers(N)    |
| + sumNaturalNumbers(N) |
| + fibonacciSeries(N)   |
| + factorial(num) : int  |
| + isPrime(num) : boolean |
| + maxOfTwo(a, b) : int  |
| + areaOfCircle(radius) : double |
| + main(args)           |
+-----+
```

5. Code

```
package studentutility;
import java.util.Scanner;

public class StudentUtilityProgram {
    // Part A: basics
    public static void displayInfo(String name, int roll, int[] marks) {
        int total = calTotal(marks);
        double percentage = calPercentage(total, marks.length);
        System.out.println("\nStudent information : ");
        System.out.println("Name of the student: " + name);
        System.out.println("Roll number: " + roll);
        System.out.println("Total marks: " + total);
        System.out.println("Percentage: " + percentage);
        displayResult(percentage);
    }

    public static int calTotal(int[] marks) {
        int total = 0;
        for (int m : marks)
            total += m;
        return total;
    }
}
```

```

public static double calPercentage(int total, int subjects) {
    return (double) total / subjects;
}

// Conditional statements
public static void evenOdd(int num) {
    System.out.println(num % 2 == 0 ? num + " is Even" : num + " is Odd");
}

public static void findLargest(int a, int b, int c) {
    int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);
    System.out.println("Largest number is: " + largest);
}

public static void displayResult(double percentage) {
    if (percentage >= 90) System.out.println("Grade: A+");
    else if (percentage >= 75) System.out.println("Grade: A");
    else if (percentage >= 60) System.out.println("Grade: B");
    else if (percentage >= 40) System.out.println("Grade: C");
    else System.out.println("Grade: Fail");
}

public static void dayOfWeek(int dayNumber) {
    switch (dayNumber) {
        case 1: System.out.println("Monday"); break;
        case 2: System.out.println("Tuesday"); break;
        case 3: System.out.println("Wednesday"); break;
        case 4: System.out.println("Thursday"); break;
        case 5: System.out.println("Friday"); break;
        case 6: System.out.println("Saturday"); break;
        case 7: System.out.println("Sunday"); break;
        default: System.out.println("Invalid day number!");
    }
}

// Loops
public static void printTable(int num) {
    System.out.println("Multiplication Table of " + num);
    for (int i = 1; i <= 10; i++)
        System.out.println(num + " x " + i + " = " + (num * i));
}

public static void displayNumbers(int N) {
    for (int i = 1; i <= N; i++) System.out.print(i + " ");
    System.out.println();
}

public static void sumNaturalNumbers(int N) {
    int sum = 0;
    for (int i = 1; i <= N; i++) sum += i;
    System.out.println("Sum of first " + N + " natural numbers = " + sum);
}

```

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public static void fibonacciSeries(int N) {
    int a = 0, b = 1;
    System.out.print("Fibonacci Series: ");
    for (int i = 1; i <= N; i++) {
        System.out.print(a + " ");
        int next = a + b;
        a = b;
        b = next;
    }
    System.out.println();
}

// Methods: with parameters and return value
public static int factorial(int num) {
    int fact = 1;
    for (int i = 1; i <= num; i++) fact *= i;
    return fact;
}

public static boolean isPrime(int num) {
    if (num <= 1) return false;
    for (int i = 2; i <= Math.sqrt(num); i++) {
        if (num % i == 0) return false;
    }
    return true;
}

public static int maxOfTwo(int a, int b) {
    return (a > b) ? a : b;
}

public static double areaOfCircle(double radius) {
    return Math.PI * radius * radius;
}

// Call each method from main
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int choice;
    do {
        System.out.println("\n--- Student Utility Program ---");
        System.out.println("1. Student Info");
        System.out.println("2. Even/Odd Check");
        System.out.println("3. Largest of Three Numbers");
        System.out.println("4. Display Grade");
        System.out.println("5. Day of Week");
        System.out.println("6. Multiplication Table");
        System.out.println("7. Display Numbers 1 to N");
        System.out.println("8. Sum of Natural Numbers");
        System.out.println("9. Fibonacci Series");
        System.out.println("10. Factorial");
        System.out.println("11. Prime Check");
        System.out.println("12. Max of Two Numbers");
    }
}

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System.out.println("13. Area of Circle");
System.out.println("0. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch (choice) {
    case 1:
        sc.nextLine();
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Roll Number: ");
        int roll = sc.nextInt();
        int[] marks = new int[3];
        System.out.println("Enter 3 subject marks:");
        for (int i = 0; i < 3; i++) marks[i] = sc.nextInt();
        displayInfo(name, roll, marks);
        break;
    case 2:
        System.out.print("Enter number: ");
        evenOdd(sc.nextInt());
        break;
    case 3:
        System.out.print("Enter three numbers: ");
        findLargest(sc.nextInt(), sc.nextInt(), sc.nextInt());
        break;
    case 4:
        System.out.print("Enter percentage: ");
        displayResult(sc.nextDouble());
        break;
    case 5:
        System.out.print("Enter day number (1-7): ");
        dayOfWeek(sc.nextInt());
        break;
    case 6:
        System.out.print("Enter number: ");
        printTable(sc.nextInt());
        break;
    case 7:
        System.out.print("Enter N: ");
        displayNumbers(sc.nextInt());
        break;
    case 8:
        System.out.print("Enter N: ");
        sumNaturalNumbers(sc.nextInt());
        break;
    case 9:
        System.out.print("Enter N: ");
        fibonacciSeries(sc.nextInt());
        break;
    case 10:
        System.out.print("Enter number: ");
        int num = sc.nextInt();
        System.out.println("Factorial = " + factorial(num));
}

```

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        break;
    case 11:
        System.out.print("Enter number: ");
        int primeNum = sc.nextInt();
        System.out.println(primeNum +
            (isPrime(primeNum) ? " is Prime" : " is Not Prime"));
        break;
    case 12:
        System.out.print("Enter two numbers: ");
        System.out.println("Max = " +
            maxOfTwo(sc.nextInt(), sc.nextInt()));
        break;
    case 13:
        System.out.print("Enter radius: ");
        System.out.println("Area = " +
            areaOfCircle(sc.nextDouble()));
        break;
    case 0:
        System.out.println("Exiting program...");
        break;
    default:
        System.out.println("Invalid choice!");
    }
} while (choice != 0);
sc.close();
}
}

```

6. Compilation & Execution Commands

```
javac StudentUtilityProgram.java
```

```
java StudentUtilityProgram
```

7. Sample Output

```
--- Student Utility Program ---
1. Student Info
2. Even/Odd Check
3. Largest of Three Numbers
4. Display Grade
5. Day of Week
6. Multiplication Table
7. Display Numbers 1 to N
8. Sum of Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Check
12. Max of Two Numbers
13. Area of Circle
0. Exit
Enter your choice: 2
Enter number: 12
12 is Even
```

8. Result

The Student Utility Program was successfully implemented and executed in Java. The program demonstrates methods, arrays, conditional statements, switch-case, loops, parameters, return values, and Scanner-based user input.